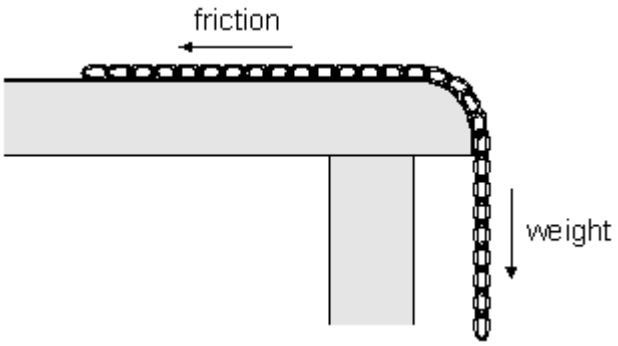
Q1. The diagram shows a chain hanging down over the edge of a table.



Two of the forces on the chain are:

- the weight of the part of the chain which is hanging over the edge;
- friction between the chain and the table.

(a) The chain is **not** moving. What does this tell you about these two forces acting on the chain?

1 mark

(b) The chain is moved slightly to the right. It begins to slide off the table.

(i) What does this tell you about these two forces now?

……..……

1 mark

(ii) Describe how the size of each force changes as the chain slides off the table.

- weight of the part of the chain hanging over the edge …….

……..…….……....

- friction between the chain and the table …….

……..…….……....

2 marks

(iii) How does the speed of the chain change as it slides off the table?

……..……

……..……

1 mark
Maximum 5 marks

Q1.

(a) they are equal

accept 'they are balanced'

1 (L5)

(b) (i) weight is greater than friction

*accept 'they are not equal **or** balanced'*

1 (L5)

(ii) it increases

1 (L6)

it decreases

1 (L6)

(iii) it increases **or** it gets faster

1 (L6)

[5]